



A multi-objective evolutionary algorithm based on dimension exploration and discrepancy evolution for UAV path planning problem

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ABSTRACT

Path planning is a crucial process for unmanned aerial vehicles (UAVs) and involves finding a path that is both short and safe. However, with the ever-increasing complexity of the environment, solving the UAV path-planning problem is challenging. Traditional path-planning methods cannot handle conflicting goals effectively, and existing objective methods lack targeted exploration mechanisms, resulting in unsatisfactory outcomes. By modeling the UAV path-planning problem via multi-objective optimization, this study designed a reasonable objective function composition for the model and considered obstacle avoidance as a hard constraint to satisfy the actual situation. A multi-objective evolutionary algorithm based on dimensional exploration and discrepancy evolution (MOEA-2DE) is presented. In particular, MOEA-2DE utilizes dimensional perturbation to identify key dimensions to facilitate prior exploration and enhance the targeted search. An adaptive evolution strategy based on population discrepancy was employed to assess the evolution process, and various methods were adopted to balance convergence and diversity. The effectiveness of the MOEA-2DE was demonstrated through the design of two intricate terrain sets and comparisons with various classic and state-of-the-art multi-objective evolutionary algorithms (MOEAs), including those designed for UAV path planning across multiple metrics. The results verify the superiority of MOEA-2DE in terms of both convergence speed and final effect.

1. Introduction

In the past few decades, unmanned aerial vehicles (UAVs) have become increasingly important in both military and civilian applications such as remote sensing, sensor data collection, UAV communication, temporary network relay, disaster monitoring, flood area monitoring, and wildfire tracking [1]. UAVs offer several advantages, such as cost-effectiveness, robustness, and the ability to execute complex tasks. When performing missions in sophisticated environments, it is crucial to ensure that UAVs follow a trajectory that keeps them away from any potential threats or restricted areas, such as high mountains or severe weather conditions. Consequently, path planning is a pivotal technology required in UAV mission planning systems, which demand a safe and effective route from the initial position to the desired destination based on specific constraints. Reasonable path planning enables UAVs to minimize their flight distance and effectively avoid potential threats.

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In the UAV path-planning problem, the complexity of finding the optimal path increases with an increase in the search space. Various methods have been developed to address this issue [2], which can be divided into five main categories. Sampling-based methods typically sample the environment as a set of nodes or other forms and then map the environment or conduct random searches to obtain feasible paths, including the rapidly exploring random tree (RRT) [3], its improved version [4], and the Voronoi method [5]. Optimal methods based on graphs generally focus on decomposition diagrams, such as Dijkstra [6], A* [7], D*, and their corresponding extensions [8]. Mathematical model-based methods primarily involve modeling the environment (kinematic constraints) and system (dynamics) and then combining cost functions with kinematic and dynamic constraint boundaries (inequalities or equations) to obtain the optimal solution. This can be achieved via approaches such as mixed integer linear programming (MILP) [9]. Evolutionary algorithms, inspired by biological behavior, skip the process of building complex environmental models and search for Pareto-optimal paths using random methods, including ant colony optimization (ACO) algorithm [10], particle swarm optimization (PSO) algorithm [11], and differential evolution (DE) algorithm [12]. Hybrid approaches, such as the hybrids of mathematical models and evolutionary algorithms [13], combine multiple methods to exploit their diverse advantages.

The optimization of UAV path planning has been proven to be nondeterministic polynomial-time (NP) complete [14]. As a widely used method for solving NP-hard problems, evolutionary algorithms can handle complex real-world scenarios that traditional modeling usually cannot solve. Population-based intelligence algorithms have been successfully employed to obtain global optimal solutions, and they have achieved promising results in various fields [15–17]. A broad variety of evolutionary algorithms have been applied to UAV path-planning problems. In the current research, the objective function mainly considers the flight distance and degree of terrain threat, while the model also needs to meet constraints such as height limits and UAV steering angle. However, some studies focused on optimizing a single objective or converting multiple objectives and constraints into a single objective optimization using the weighted sum method. In this case, the adjustment of weights has a significant impact, and the selection of suitable objective requirements becomes infeasible. Therefore, multi-objective evolutionary algorithms (MOEAs) have been developed to address UAV path-planning problems that compromise multiple objective conflicts. However, existing MOEAs for UAV path planning rely primarily on large-scale random searches, which require a large number of evaluations and a considerable amount of time, leading to unique results within limited evaluation and time constraints.

To describe the UAV path-planning problem accurately, this study modeled it as a multi-objective problem, considering the actual needs of UAV flights with objective functions composed of flight distance and altitude. Terrain threats are treated as hard rather than soft constraints, which are more consistent with real-world scenarios. The model also incorporates UAV mobility constraints. To efficiently obtain high-quality solutions, a MOEA based on dimension exploration and discrepancy evolution (MOEA-2DE) was proposed. The algorithm consists of two strategies: a key dimension exploration strategy and an adaptive evolution strategy based on population discrepancy. The key dimension exploration strategy partitions the search space into dimensions and perturbs each dimension to identify the key dimensions that significantly affect the current problem so as to conduct targeted optimization and exploit greater possibilities. The adaptive evolution strategy based on population discrepancy adopts the Kullback-Leibler divergence (KLD) [18] to measure the intergenerational distribution differences among populations and assesses the current evolution trend based on the obtained differences. Various offspring generation methods have been employed to adapt to different trends in the evolutionary process, thereby ensuring a balance between convergence and diversity of solutions. To validate the performance of MOEA-2DE, two sets of complex terrain problems were created. The proposed algorithm was compared with several classical and latest algorithms, including both theoretical algorithms and algorithms specifically designed for UAV path planning. Multiple metrics were employed for the evaluation. These results demonstrate the superiority of MOEA-2DE. The contributions of this article are as follows:

- A UAV path-planning model was constructed to consider flight distance and altitude as objective functions and to incorporate terrain threats into hard constraints. The objective function composition was described based on an actual situation with reasonable explanations.
- A MOEA based on dimension exploration and discrepancy evolution was proposed, which involves a key dimension exploration strategy and an adaptive evolution strategy based on population discrepancy, providing a key dimension search and optimizing the evolutionary processes to quickly obtain high-quality solutions.
- Two complex terrain problem sets were created to verify the effectiveness of the proposed algorithm against various algorithms.

The remainder of this paper is organized as follows. Section 2 provides an overview of the preliminaries and literature review. The UAV path-planning model and the proposed algorithm are described in Section 3. The experimental settings and analyses are presented in Section 4. Finally, Section 5 concludes the paper and outlines future research directions.

2. Preliminaries and literature review

2.1. Multi-objective optimization

Multi-objective optimization aims to deal with multiple conflicting objectives simultaneously and has been applied to numerous practical problems such as resource allocation, transportation system decision-making, and robot gripper optimization [19]. Without losing generality, a multi-objective optimization problem can be expressed as [20]

$$\begin{aligned} \min \mathbf{F}(\mathbf{x}) &= \min (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_M(\mathbf{x})) \\ \text{subject to } &: \mathbf{x} \in \mathbf{R}^D, \end{aligned} \quad (1)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_D)$ denotes the decision variable with D -dimension in $\mathbf{R}$, $f(\mathbf{x})$ represents the single objective function, and M is the number of objectives. The Pareto-dominance relationship can be used to determine the better solution under multiple objectives. If and only if $\forall i \in \{1, 2, \dots, M\}, f_i(\mathbf{x}_1) \leq f_i(\mathbf{x}_2)$ and $\exists j \in \{1, 2, \dots, M\}, f_j(\mathbf{x}_1) < f_j(\mathbf{x}_2)$ are satisfied, solution $\mathbf{x}_1$ is said to dominate solution $\mathbf{x}_2$. A Pareto-optimal solution is one that is not dominated by the other solutions. The set of Pareto-optimal solutions is called Pareto-optimal set (PS). The objective space corresponding to PS is called Pareto-optimal front (PF).

2.2. Related work

In recent years, evolutionary algorithms used to solve UAV path-planning problems have attracted increasing attention. The objectives of the UAV path-planning problem are diverse but commonly include minimizing the path length and reducing the threat level. When dealing with multiple objectives simultaneously, modeling as a single objective or multi-objective optimization problem involves two different means of settling. Furthermore, various constraints must be considered, such as the limitations of the aircraft turning angle and other practical requirements. These constraints play a crucial role in ensuring the feasibility and applicability of path-planning solutions.

When modeling as a single objective optimization problem, the primary goal is to minimize the path length. For this condition, the objective of the model built by Yu et al. [21] was a shorter path length, whereas the threat level and limits on the turning angle were expressed as constraints to ensure the feasibility of the solutions. To find the solution, an improved grey wolf optimization (GWO) algorithm, HGWODE, was proposed, which combines the DE operator to balance the capabilities of exploration and exploitation.

When integrating multiple-objective decisions into a single objective problem, the common approach adopted in the literature is the weighted sum method to achieve a combination of multiple objectives and constraints. Based on whether the objectives and constraints are processed separately, these studies can be classified into two types: separable and nonseparable. In the separable type, the evaluation function consists of specific objectives without constraints that serve as additional conditions for filtering the solutions. For example, Qu et al. [22] combined fuel and threat costs into a single function and proposed a hybrid algorithm by merging a simplified grey wolf optimizer (SGWO) and a modified symbolic organization search (MSOS) to address the problem. Modeled as a constrained objective optimization problem, the objective function constructed by Yu et al. [23] was formed by the weighted sum of the flight distance and threat level, and feasible solutions were selected with lower constraint values. To address this issue, this work designed an adaptive-selection mutation-based constrained DE algorithm that selects preferred individuals to participate in more mutations. Compared to the separable type, the literature on the non-separable type adds constraints to the evaluation function and incorporates them into the objectives. This approach increased the driving force required to satisfy these constraints. Phung et al. [24] considered the weighted path length, threat, path smoothness, and flight altitude as components of a single objective optimization problem and proposed a new PSO algorithm based on spherical vectors (SPSO) to search the configuration space effectively. By combining multiple threats, path length, flight altitude, and climb constraints into a single objective function, Chai et al. [12] introduced a multi-strategy fusion DE (MSFDE) algorithm to handle the issue. Similar to Chai's model, Liu et al. [25] proposed a multi-strategy GWO algorithm (AS-GWO) that integrates nonlinear convergence factors, adaptive inertia weights, and spiral functions for agricultural UAV trajectory planning. Similarly, Huang et al. [26] considered the cost of path length, flight altitude, threat, and physical constraints and formed an evaluation function by adding weights to each portion. A novel and efficient PSO algorithm (ACVDEPSO) that integrates a cylindrical coordinate vector, DE, and an adaptive parameter strategy is proposed to alleviate premature convergence. Similar literature comprises the majority of research on UAV path-planning problems, which can be found in the cited articles [27–29].

Modeling the UAV path-planning problem as a single objective optimization problem is conducive to simplifying the solution process. However, the modeling method of a single objective does not consider multiple factors because the single objective optimization problem formed by the combination of multiple objectives obscures the optimal selection of objectives. The weight-sum-based method also exhibits stringent requirements for parameter settings [30], which vary depending on the problem. Consequently, research has been conducted on modeling the UAV path-planning problem as a multi-objective optimization problem to balance multiple goals and provide compromise solutions. Most of these related studies focused on forming functions for each objective separately and solving multiple objective functions simultaneously, with constraints used to screen the feasibility of the solutions. For instance, the multi-objective model proposed by Wan et al. [31] involved a shorter path length and a lower threat degree. Therefore, an accurate path-planning method for UAVs (APPMS) based on an enhanced multi-objective swarm intelligence algorithm was proposed. The precise swarm intelligence search method based on the improved ACO strengthens global and local search capabilities using preferred search directions and random neighborhood search mechanisms. Similarly, Peng et al. [30] introduced a constrained MOEA based on decomposition (M2M-DW), which contains a locally infeasible utilization mechanism. Yu et al. [32] emphasized that the knee point can provide abundant geometric information and proposed a DE algorithm based on the knee point (DEAKP), where the knee point is determined according to the minimum Manhattan distance to guide the search direction. Unlike the aforementioned goals, Nikolas et al. [33] considered energy consumption and noise as optimization objectives and suggested a hybrid approach, blending a conventional version of an evolution strategy (ES), non-dominated sorting genetic algorithm II (NSGAI), multi-objective covariance matrix adaptation evolution strategy (MO-CMA-ES), and a gradient-based limited-memory Broyden-Fletcher-Goldfarb-Shanno algorithm with a bounds (L-BFGS-B) approach. Zhang et al. [34] regarded the weighted sum of objectives as one of the optimization functions and constraints as another optimization function to form a multi-objective optimization problem. They designed a multi-objective

PSO (MOPSO) algorithm with multi-mode collaboration based on reinforcement learning (MCMOPSO-RL). Three update modes were incorporated and calculated using cumulative performance to achieve a comprehensive search.

Overall, compared with the single objective optimization problem, modeling as a multi-objective optimization problem shows obvious superiority by considering the selectivity and contradiction of the objectives. However, existing MOEAs for UAV path-planning problems lack the ability to target searches, which leads to a large search cost in exchange for convergence, resulting in undesirable effects with limited resources. The algorithm proposed in this study aims to enhance targeted search to solve the above issues. Additionally, strengthening the utilization of historical experience to further assist in a targeted search is an important component of the proposed algorithm.

3. UAV path-planning model and the proposed algorithm

In this section, a multi-objective optimization model for UAV path planning is described along with path representation and the definition of objective functions and constraints. Next, we present the design of an evolutionary algorithm based on dimension exploration and discrepancy evolution, which includes a key dimension exploration strategy and an adaptive evolution strategy based on population discrepancy. The algorithm is expected to fully explore the search space to determine the optimal global path that satisfies these constraints.

3.1. UAV path-planning model

3.1.1. Multi-objective modeling for UAV path-planning problem

This section presents a multi-objective model for the UAV path-planning problem. From a practical perspective, the primary goal of the UAV path-planning problem is to determine the shortest path, as defined in Eq. (2). Parameter n denotes the number of trajectory points, and $l_{i,i-1}$ represents the distance between the i -th and $(i-1)$ -th point. Because low-altitude flight is ideal for both actual UAV flights and military missions, flight altitude is considered another objective function for reducing the risk of exposure, as represented by Eq. (3). H indicates the height of flight.

$$f_1 = \sum_{i=2}^n l_{i,i-1} \quad (2)$$

$$f_2 = \sum_{i=1}^n H_i \quad (3)$$

Notably, these two goals, shortest path and low-altitude, often conflict in real-life scenarios. Obstacles may exist along the shortest path of the drone, leaving two options. The first option is to increase flight altitude to overcome obstacles; however, the cost of such a situation is greater energy consumption and higher exposure risk. Otherwise, it would be necessary to determine an alternative path. At this point, the selected path may not be the shortest path; however, sacrificing a portion of the shortest path to determine a path with lower energy consumption and risk is desirable. Therefore, this study asserts that the goals of finding a shorter path and maintaining a lower flight altitude constitute a conflicting multi-objective optimization problem.

3.1.2. Constraints of the model

In addition to defining the objective functions, it is important to appropriately define flight constraints. Unlike other studies that incorporated threats into the objective function, this study suggests that obstacle avoidance should be considered a strict rather than a flexible constraint. Maintaining a safe distance from the terrain is crucial to obtaining a feasible solution. Specifically, each trajectory point must be at a higher altitude than the nearby terrain, that is,

$$H_i \geq \max_{p \in P} (H_p) + h_{safe}, \quad (4)$$

where P indicates the nearest terrain point set and h_{safe} denotes the safe distance constant.

In addition, the model defines the mobility constraints of the UAVs, such as the horizontal and vertical steering angles. The horizontal steering angle θ_h is calculated by the angle between consecutive path points and should not exceed the maximum range θ_{hmax} , as defined:

$$\theta_h = \cos^{-1} \left(\frac{\mathbf{L}_{i,i-1} \cdot \mathbf{L}_{i+1,i}}{\|\mathbf{L}_{i,i-1}\| \|\mathbf{L}_{i+1,i}\|} \right) \leq \theta_{hmax}, \quad (5)$$

where $\mathbf{L}_{i,i-1}$ denotes the vector from the i -th to $(i-1)$ -th point, and $\mathbf{L}_{i+1,i}$ refers to the same indication with a different point vector. Similarly, the vertical steering angle θ_v is obtained by the angle of the height difference between the path points, which also has a maximum range θ_{vmax} , given as

$$\theta_v = \tan^{-1} \left(\frac{\Delta H}{\Delta d} \right) \leq \theta_{vmax}, \quad (6)$$

where ΔH represents the height difference between two trajectory points and Δd indicates the horizontal projection distance between two trajectory points. When a constraint is violated, it is necessary to adjust the part that violates the constraint, including

its position and flight altitude. In this study, a random adjustment operation was adopted for the constraints violations. The entire mathematical model was developed by combining Eq. (2) to Eq. (6).

3.1.3. Path-smoothing method

The path of a UAV comprises a series of waypoints that serve as decision variables for the model. To obtain a smooth and continuous path, path-smoothing methods are employed in the UAV path-planning problem to transform a set of waypoints into a smooth set of trajectory points. The B-spline method is widely used as a path-smoothing method [23,30] owing to its simple and flexible ability to characterize paths and was used in this study. Supposing that there are $D + 1$ waypoints with coordinates $(x_w^0, y_w^0, z_w^0), \dots, (x_w^D, y_w^D, z_w^D)$ to form a curve of n trajectory points (x, y, z) , thus,

$$\begin{cases} x^t = \sum_{i=0}^D x_w^i \cdot B_{i,j}(t), \\ y^t = \sum_{i=0}^D y_w^i \cdot B_{i,j}(t), \\ z^t = \sum_{i=0}^D z_w^i \cdot B_{i,j}(t), \end{cases} \quad (7)$$

where $B_{i,j}(t)$ is the blending function, j denotes the order of the curve, and t is a parameter with a fixed change in the step size from $[0, D + 1 + j]$. To obtain the blending function, it is necessary to obtain the knot vector as follows:

$$\begin{cases} \text{Knot}(i) = 0, & \text{if } i < j, \\ \text{Knot}(i) = i - j, & \text{if } j \leq i \leq n - 1, \\ \text{Knot}(i) = n - j, & \text{if } n \leq i. \end{cases} \quad (8)$$

Hence, the blending function can be calculated recursively as follows,

$$B_{i,1}(t) = \begin{cases} 1, & \text{Knot}(i) \leq t < \text{Knot}(i + 1), \\ 0, & \text{otherwise,} \end{cases} \quad (9)$$

$$B_{i,j}(t) = \frac{(t - \text{Knot}(i)) \times B_{i,j-1}(t)}{\text{Knot}(i + j - 1) - \text{Knot}(i)} + \frac{(\text{Knot}(i + j) - t) \times B_{i+1,j-1}(t)}{\text{Knot}(i + j) - \text{Knot}(i + 1)}. \quad (10)$$

At this point, the B-spline method was used to smooth the path into a trajectory.

3.2. Proposed algorithm

3.2.1. General framework

To solve the UAV path-planning problem adaptively, the encoding method of individuals tends to reflect changes in paths. In this study, each individual's decision variable represented a path, specifically a matrix of $n \times 3$, where n denotes the number of waypoints. Compared to existing coding methods, such as defining the entire path as a vector for decision variables, the proposed encoding method significantly reduces the number of dimensions, thus improving the efficiency and accuracy of the evolution.

Algorithm 1 The procedure of the proposed algorithm.

Input: N (population size), n (the number of waypoints)

Output: A series of optimal solutions

```

1: Initialize the population  $P$  that meets the constraints and evaluate it based on the objective function;
2: Perform key dimension exploration strategy to obtain key dimension set  $Dim$ ;  $\leftarrow$  Algorithm 2
3: while the termination conditions are not met do
4:   for  $i = 1 \rightarrow N/2$  do
5:     if Dimension disturbance sign  $S$  is true then
6:       Select two elite individuals as the parents;
7:       Perform the offspring generation method of exploring key dimensions on selected parents to generate two offspring  $o_1$  and  $o_2$ ;
8:     else
9:       Randomly choose two individuals as parents;
10:      Generate two offspring  $o_1$  and  $o_2$  via binary crossover on selected parents;
11:    end if
12:    Constraint detection and adjustment on  $o_1$  and  $o_2$ ;
13:    Evaluate  $o_1, o_2$  and add them into the offspring population  $O$ ;
14:  end for
15:  Apply environmental selection to  $P \cup O$ ;
16:  Update  $S$  by adaptive evolution strategy based on population discrepancy;
17: end while

```

The pseudocode of the proposed algorithm is provided in Algorithm 1, which outlines the key component operations, including the two strategies. First, population P is equipped with n individuals, each of which must satisfy the constraints and obtain the evaluation values of the objective functions (Line 1). Next, a key dimension exploration strategy is conducted to obtain the key dimensions Dim that affect the current problem (Line 2), which are then employed in the subsequent strategy. The iterations were continued until the termination conditions were satisfied. In each iteration, it is first determined whether the dimensional perturbation sign S is TRUE

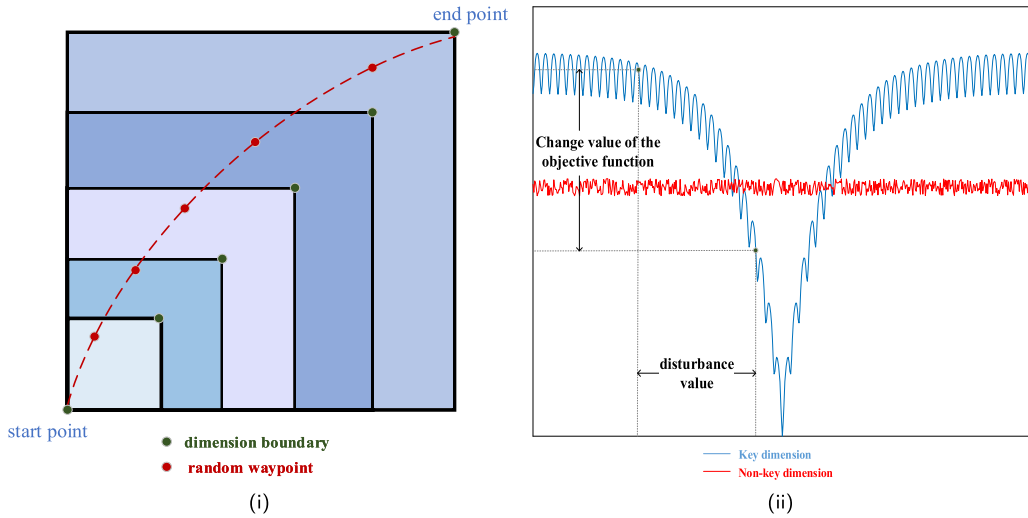


Fig. 1. Intuition behind the idea of the key dimension exploration strategy. (i) Schematic of search space division. (ii) Schematic of dimension perturbation.

(Line 4). If it is TRUE, elite individuals are selected as parents, and the offspring generation method for exploring key dimensions is applied to generate the offspring (Lines 6-7). Otherwise, the parents are randomly chosen to generate offspring via binary crossover (Lines 9-10). The generated offspring must verify the constraints and adjust parts that violate them (Line 12). Subsequently, the objective function values of each offspring are evaluated, and the offspring are incorporated into offspring population O (Line 13). Finally, the parent population P and offspring population O are merged for environmental selection to obtain a new generation of the population (Line 15). Accordingly, an adaptive evolution strategy based on population discrepancy is employed to update the dimensional perturbation sign S to balance the convergence and diversity of the next generation (Line 16). Once the termination condition is satisfied, a series of Pareto-optimal solutions are obtained. Further details of the proposed strategies are provided in the following subsections.

3.2.2. Key dimension exploration strategy

In the UAV path-planning problem, drones must navigate through various terrain environments that vary significantly across different locations. For example, when a drone flies over a mountainous forest, it can either increase its flight altitude or detour around its peak. These two circumstances present significant differences in goals owing to vastly different flight heights and path lengths. Naturally, there are remarkable differences in the impact of the waypoints within this range. Because the decision variables in this problem are represented as drone waypoints, any violent changes in the terrain directly affect these variables, consequently affecting the overall objective. Investigating and analyzing the range that is susceptible to fluctuation can not only accelerate convergence but also facilitate the exploration of greater possibilities for obtaining better paths. Motivated by the abovementioned facts, a key dimension exploration strategy is proposed to achieve a priori exploration of the problem, identify the key dimensions that influence the problem, and conduct targeted searches.

Fig. 1 illustrates the intuition behind the key dimension exploration strategy, which consists of two parts: division of the search space and dimension perturbation. In Fig. 1(i), the entire picture represents the search space that encompasses the area that the drone must traverse. The decision variable was composed of a series of waypoints, with each one-dimensional decision variable indicating the waypoint to be searched. The number of waypoints and size of the flight space determine the search space for each one-dimensional decision variable. For example, if the flight space is a two-dimensional map of 200×200 pixels with 50 waypoints, the size of the search space for each dimension is 4×4 pixels. Each dimension is characterized by a defined boundary represented by solid green dots. The red dots signify random waypoints within the corresponding boundaries that can form flight paths using smoothing methods. Once the search space is divided, perturbing each dimension is essential to identifying the key dimensions. Fig. 1(ii) shows a graphical representation of the dimensional perturbations. The blue and red curves denote the different dimensions. When a disturbance was applied, the red curve underwent slight changes, whereas the blue curve exhibited significant alterations, which were assumed to be the easily affected dimensions. Consequently, the dimension represented by the blue curve is marked as the key dimension, whereas the red curve represents the nonkey dimension.

The pseudocode for the key dimension exploration method is presented in Algorithm 2. Based on one parent, only one dimension is disturbed at a time (Line 2). The size of the disturbance should not be too small to avoid filtering all the dimensions. Following each one-dimensional disturbance, it was necessary to compare and record the disturbance results to observe significant changes (Line 5). Consequently, the degree of change was reflected in the differences in the objective functions. The key dimensions were ultimately identified as those with differences greater than the mean (Line 7). This implies that these dimensions are more susceptible to fluctuations at the average level and can emerge as greater exploration possibilities.

Algorithm 2 The procedure of key dimension exploration strategy.**Input:** D (dimension size)**Output:** Dim (key dimension set)

- 1: Divide the search space for each dimension;
- 2: Select a parent p as the benchmark for the disturbance;
- 3: **for** $i = 1 \rightarrow D$ **do**
- 4: Disturb the i -th dimension of p and evaluate its objective function value;
- 5: Record the difference $Diff$ between each dimensional objective function value after disturbance and that of p ;
- 6: **end for**
- 7: Calculate the mean value of the difference $Diff$;
- 8: Dimensions with a difference greater than the mean value are merged into the key dimension set Dim ;

3.2.3. Adaptive evolution strategy based on population discrepancy

These algorithms are expected to yield a solution set with strong convergence and wide diversity. Strong convergence indicates that the solution supports the convergence of the objective function, whereas better diversity reflects a solution set with a wide distribution of solutions, showing significant differences and selectivity. Balancing convergence and diversity is essential to obtaining high-quality solutions. Intuitively, analyzing the distribution among individuals can reflect the distribution trend of the current population and, to some extent, represent the diversity of the solution set. Inspired by this, an adaptive evolution strategy based on population discrepancy was proposed, which leverages KLD to calculate the population differences between generations and adjusts the measures of offspring generation suitable for the current evolution trend according to the changes to realize a balance between convergence and diversity.

KLD [18] is a measure of the distance between two distributions and is widely used in fields such as pattern recognition [35] and transfer learning [36]. Here, KLD is used to measure the population differences between generations:

$$D_i(P_{i-1} \| P_i) = \sum_{x=1} P_{i-1}(x) \log \frac{P_{i-1}(x)}{P_i(x)}, \quad (11)$$

where i represents generation and P denotes population. The calculated distribution differences reflect current evolutionary trends. If the distribution difference increases compared to the previous ones ($Diff_i \geq Diff_{i-1}$), it signifies that the distribution differences of the population are relatively large, and convergence needs to be strengthened at this time. Conversely, if the distribution difference decreases ($Diff_i < Diff_{i-1}$), the diversity of the offspring must increase. Therefore, in cases of significant differences, the offspring generation method, which involves exploring key dimensions with elite individuals, can enhance the investigation of elite individuals and facilitate the search for better solutions to promote convergence. In contrast, the high-probability crossover method of random parents was applied to increase individual differences and avoid long-term engagement with the local optimum.

The key dimension exploration method based on elite individuals was used for their respective key dimensions (obtained from the key dimension exploration strategy). Specifically, the selected parents were elite individuals from the first non-dominated level of environmental selection. According to the dimension division, the selected key dimensions of the parent were randomly searched within the range to generate the offspring, that is,

$$v = rand \cdot (U - L) + L, \quad (12)$$

$$y_i = \begin{cases} v_i, & \text{rand} \leq SD \text{ or } i \in Dim, \\ x_i, & \text{otherwise,} \end{cases} \quad j = 1, 2, \dots, D, \quad (13)$$

where v is the newly generated vector in the search space and $rand$ denotes the [0,1] random matrix. U and L represent the upper and lower bounds of the dimensions, respectively. In Eq. (13), y denotes the decision variable vector of the generated offspring, and x represents the decision variable vector of the selected parent. SD indicates the mutation probability of other dimensions that are not key dimensions, and D is the dimension size. When a high-probability crossover method based on random parents is selected, binary crossover and single-point mutation methods are used to generate offspring. Two parents were selected randomly from the entire population. The selected dimensions of the decision variable vector, determined by probability, were exchanged for the selected parents, and a single point was mutated to form new offspring. Fig. 2 illustrates the design concept of the adaptive evolution strategy based on population discrepancy.

3.2.4. Time complexity analysis

The proposed algorithm comprises two main strategies. The calculation of the key dimension exploration strategy is based on dimensional exploration with a complexity of $O(D)$, where D denotes the dimension size, that is, the number of waypoints. Second, the complexity of the KLD calculation for the adaptive evolution strategy based on population discrepancy is $O(DN)$, where N represents the population size. As a universal part, the generation of offspring in each iteration requires a time complexity of $O(DN)$. For the UAV path-planning problem, the complexity of one evaluation is $O(n)$, where n is the number of trajectory points. Consequently, the evaluation complexity per generation is $O(Nn)$. Environmental selection involves a time complexity of $O(MN^2)$ and $O(MN \log(N))$ for non-dominated sorting and crowding distance computation [37], respectively, where M is the number of objectives. In conclusion, the computational complexity of MOEA-2DE is $O(\max(DN, Nn, MN^2))$; that is, it depends on the number of waypoints and trajectory points, population size, and number of objectives.

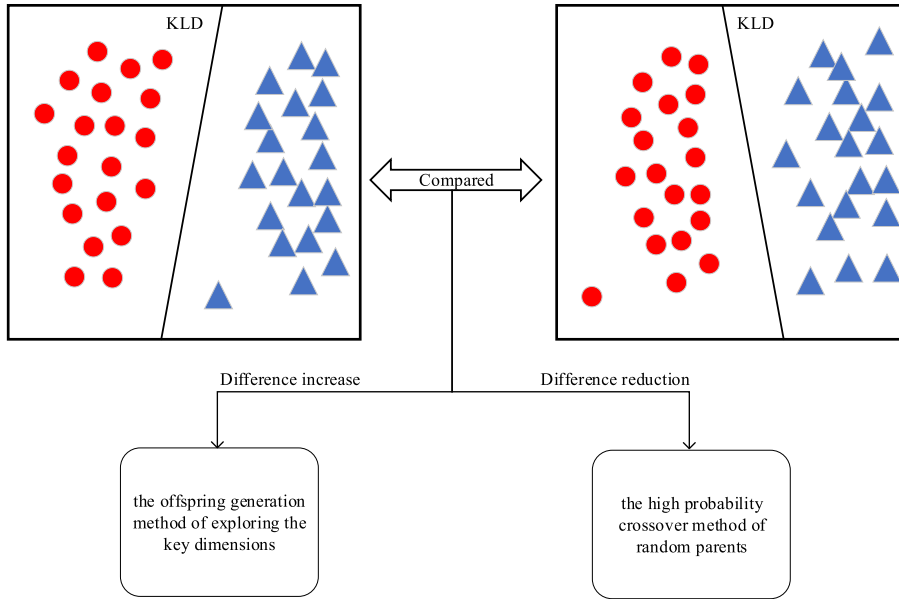


Fig. 2. Schematic of the adaptive evolution strategy based on population discrepancy.

Table 1
Parameter settings for terrain set.

	γ_1	γ_2	γ_3	γ_4	γ_5	γ_6	γ_7
peak	3π	1/10	9/10	1/2	1/2	3/10	—
rugged1	1	2	0.3	0	0.2	2	0.1
rugged2	5	5	1.2	1	3	1.8	1

4. Experiments and analysis

Comparative experiments were conducted to verify the effectiveness of MOEA-2DE. To solve the UAV path-planning problem, two sets of terrains were constructed considering the characteristics of complex environments, including peak and rugged terrain scenes. Multiple metrics were used to measure the performance of various comparison algorithms. The sensitivity of the parameter and running time were also investigated. A detailed analysis and discussion are provided based on these results.

4.1. Terrain sets and parameter settings

The simulations were executed using MATLAB on a server equipped with a 2.5 GHz CPU and 32.0 GB of RAM. Terrain sets were constructed using simulated digital terrain and divided into peak and rugged terrains based on the classification of the actual terrain. Peak terrain simulates dense mountainous terrains, requiring drones to find shorter paths to avoid collisions. Two terrain test problems were designed for the peak terrain, both of which posed threats along the shortest path of the UAV flight. Peak2 was more complex than peak1, as it included more threats and narrower spaces. The peak terrain set is constructed by combining the basic terrain formed by Eq. (14) with the peaks formed by Eq. (15) [31]. The parameters x and y represent the coordinates of the horizontal plane, whereas the other parameters are listed in Table 1, following the work of Wan et al. [31]. The peak settings described in Eq. (15) are listed in Table 2, where k denotes the number of peaks for different problems. The data in Table 2 correspond to the peaks of the peak terrain originally created based on the principles of simplicity and complexity.

$$z_1(x, y) = \sin(y + \gamma_1) + \gamma_2 \sin(x) + \gamma_3 \cos(\gamma_4 \sqrt{y^2 + x^2}) + \gamma_5 \sin(\gamma_5 \sqrt{y^2 + x^2}) + \gamma_6 \cos(y) \quad (14)$$

$$z_2(x, y) = \sum_{i=1}^k H(i) \times \exp\left(-\left(x - X_c(i)\right)^2 / X_t(i) - \left(y - Y_c(i)\right)^2 / Y_t(i)\right) \quad (15)$$

The rugged terrain simulated a wide but uneven outdoor terrain with continuous high and low depressions encountered during the flight. Similarly, two terrain test problems, rugged1 and rugged2, were developed for rugged terrains. Rugged2 is more sophisticated than rugged1, featuring greater magnitude and greater undulation. Rugged terrain was constructed using Eq. (16) [30], whose parameters were adjusted to form rugged1 and rugged2, as listed in Table 1, based on the work of Peng et al. [30]

Table 2
Parameter settings for peak terrains.

peakTerrain	peakNumber	(x_c, y_c)	(x_t, y_t)	H (km)
peak1	1	(50,70)	(80,50)	2.3
	2	(80,180)	(300,220)	2.5
	3	(150,90)	(120,200)	3
	4	(150,145)	(120,100)	3.5
peak2	1	(119,100)	(100,50)	2
	2	(30,70)	(80,50)	2.3
	3	(60,50)	(100,80)	3
	4	(50,150)	(150,200)	3.2
	5	(60,100)	(300,220)	3.4
	6	(150,90)	(120,250)	3.6
	7	(100,160)	(250,160)	4.8
	8	(140,40)	(200,50)	5.3
	9	(100,180)	(230,50)	3.8
	10	(90,70)	(150,100)	4.5
	11	(150,150)	(90,60)	2.8

Table 3
Parameter setting for the experiment.

Parameter	value
Population size	20
The total number of evaluations	4,000
The number of waypoints for peak terrains	50
The number of waypoints for rugged terrains	10
The number of trajectory points for peak terrains	200
The number of trajectory points for rugged terrains	50
h_{safe} for peak terrains	200
h_{safe} for rugged terrains	2
Crossover probability η_c	0.9
Mutation probability η_m	0.1
CR for DE operator	0.3
F for DE operator	0.3
SD for MOEA-2DE	0.3

$$z3(x, y) = \sin(y + \gamma_1) + \gamma_2 \sin(x) + \gamma_3 \cos(y) + \gamma_4 \cos(y) + \gamma_5 \cos(\gamma_6 \sqrt{x^2 + y^2}) + \gamma_7 \sin(\gamma_7 \sqrt{x^2 + y^2}) \quad (16)$$

The basic parameter settings used in the experiments are listed in Table 3. The peculiar parameters in the compared algorithms were consistent with those in the original studies.

4.2. Compared algorithms and performance metrics

The comparison algorithms included a variety of classic MOEAs and advanced improved algorithms. In addition, the proposed algorithm was compared with newly proposed MOEAs applied to the UAV path-planning problem. Classic MOEAs include NSGAI [37], MOEA/D [38], MOPSO [39], and MODE [40], whereas the developed algorithms include NSGAI/SDR [41], MOEA/D-DU [42], CMOPSO [43], LMPFE [44], MOEA/D-UR [45], and PREA [46]. The code for several comparative algorithms was derived from PlatEMO [47] and adapted for the proposed project.

NSGAI is a widely used dominance-based evolutionary algorithm with numerous improved versions. Based on the modification of NSGAI, NSGAI/SDR introduces a novel relationship that incorporates an adaptive niching technology for measurement. MOEA/D is a classical decomposition-based evolutionary algorithm that serves as the foundation for all decomposition-based evolutionary algorithms. MOEA/D-DU implements MOEA/D by utilizing the vertical distance from the solution to the weight vector in the objective space to improve performance. Another MOEA/D-based algorithm, MOEA/D-UR, designs detection metrics to determine when to adjust weights and partition the objective space to realize an adaptive weight adjustment. MOPSO is a basic representation of a multi-objective PSO algorithm, whereas CMOPSO is based on a competition mechanism. MODE is a MOEA that incorporates a DE operator. With respect to state-of-the-art MOEAs, LMPFE refers to a local model-based PF estimation method for estimating the geometric structure of a PF during its evolution. PREA employs a ratio-based indicator with an infinite norm to evaluate the fitness values of each individual in the current population and chooses the solution with the best fitness value to identify a promising region in the objective space. Among the MOEAs applied to UAV path planning, APPMS [31], DEAKP [32], and MCMOPSO-RL [34] provide good performance and are suitable for comparison. Their introductions are presented in the Related Works section.

Because the PF of the designed UAV path-planning problem is unknown, the performance of the algorithms was quantified using hypervolume (HV) [48], a commonly used multi-objective metric defined in Eq. (17):

Table 4

Averaged HV value, PD value and corresponding standard deviations obtained by APPMS, MODE, MOPSO, CMOPSO, NSGAI, NSGAI/SDR, MOEA/D, MOEA/D-DU, LMPFE, MOEA/D-UR, PREA, MCMOPSO-RL, DEAKP and MOEA-2DE on peak terrains.

Terrain	Metrics	peak1		peak2	
		HV	PD	HV	PD
Algorithms					
APPMS		3.0417E-01(+) $\pm 9.27E-02$	2.7366E+03(+) $\pm 6.71E+02$	3.4552E-01(+) $\pm 8.28E-02$	2.8587E+03(+) $\pm 1.16E+03$
MODE		2.6705E-01(+) $\pm 7.88E-02$	1.8868E+03(+) $\pm 4.38E+02$	3.0899E-01(+) $\pm 9.05E-02$	1.1954E+03(+) $\pm 3.02E+02$
MOPSO		4.7245E-01(+) $\pm 9.56E-02$	3.0525E+03(+) $\pm 7.07E+02$	4.0649E-01(+) $\pm 9.56E-02$	2.1697E+03(+) $\pm 6.57E+02$
CMOPSO		5.2401E-01(+) $\pm 5.15E-02$	2.0459E+03(+) $\pm 4.53E+02$	4.6503E-01(+) $\pm 6.35E-02$	1.4889E+03(+) $\pm 3.05E+02$
NSGAI		5.3016E-01(+) $\pm 5.24E-02$	2.2510E+03(+) $\pm 6.84E+02$	4.9597E-01(+) $\pm 7.45E-02$	1.6375E+03(+) $\pm 4.07E+02$
NSGAI/SDR		5.4062E-01(+) $\pm 5.06E-02$	4.1454E+03(+) $\pm 9.15E+02$	5.0613E-01(+) $\pm 6.12E-02$	3.1387E+03(+) $\pm 7.61E+02$
MOEA/D		5.2354E-01(+) $\pm 3.52E-02$	2.2110E+03(+) $\pm 5.62E+02$	4.8905E-01(+) $\pm 5.70E-02$	1.5020E+03(+) $\pm 4.53E+02$
MOEA/D-DU		3.0210E-01(+) $\pm 1.04E-01$	4.2521E+03(+) $\pm 1.05E+03$	2.5877E-01(+) $\pm 8.99E-02$	2.2682E+03(+) $\pm 9.08E+02$
LMPFE		5.4931E-01(+) $\pm 3.09E-02$	2.4741E+03(+) $\pm 5.37E+02$	4.9834E-01(+) $\pm 6.25E-02$	1.6196E+03(+) $\pm 4.21E+02$
MOEA/D-UR		4.8884E-01(+) $\pm 7.79E-02$	3.1845E+03(+) $\pm 7.66E+02$	4.3142E-01(+) $\pm 7.19E-02$	2.0676E+03(+) $\pm 6.01E+02$
PREA		5.4111E-01(+) $\pm 5.39E-02$	2.5007E+03(+) $\pm 8.33E+02$	4.6733E-01(+) $\pm 7.63E-02$	1.3815E+03(+) $\pm 3.62E+02$
MCMOPSO-RL		5.7398E-01(+) $\pm 5.77E-02$	5.2516E+03(+) $\pm 9.87E+02$	4.9936E-01($\approx$) $\pm 1.19E-01$	3.5263E+03($\approx$) $\pm 9.02E+02$
DEAKP		2.1178E-01(+) $\pm 6.88E-02$	2.3081E+03(+) $\pm 5.62E+02$	2.2392E-01(+) $\pm 4.88E-02$	1.3658E+03(+) $\pm 3.47E+02$
MOEA-2DE		6.0578E-01 $\pm 2.55E-02$	6.6341E+03 $\pm 6.38E+02$	5.4174E-01 $\pm 3.69E-02$	3.3988E+03 $\pm 5.65E+02$

$$HV(A, \mathbf{r}) = \mathcal{L} \left(\bigcup_{\mathbf{a} \in A} \{\mathbf{b} \mid \mathbf{a} \geq \mathbf{b} \geq \mathbf{r}\} \right), \quad (17)$$

where A denotes the solution set, $\mathbf{r}$ is a reference point, and $\mathcal{L}$ represents the Lebesgue measure of a set. The final solution set was normalized when calculating the HV because of the different scaled objective values, and the reference point was set to $[1, 1]$ in this study. A higher HV value indicates a stronger convergence and diversity of the solution, demonstrating better performance. Although the HV can consider both, it is difficult to single out the diversity performance from these values as a mixed indicator. In the UAV path-planning problem, conflicts exist in objective functions composed of shorter paths and lower flight heights, and the selectivity for different solutions can be evaluated based on diversity. As an indicator of diversity, pure diversity (PD) [49] can measure diversity independently of any parameter or reference, thereby mitigating a reduction in objectivity. As a result, PD was added as an indicator to evaluate diversity, as defined in Eq. (18). The larger the PD value, the better the diversity of the entire solution set.

$$\begin{aligned} PD(A) &= \max_{a_i \in A} (PD(A - a_i) + d(a_i, A - a_i)) \\ d(a, A) &= \min_{a_i \in A} (\text{dissimilarity}(a, a_i)) \end{aligned} \quad (18)$$

4.3. Results on peak terrain scene

The average HV, PD, and corresponding standard deviations obtained by APPMS, MODE, MOPSO, CMOPSO, NSGAI, NSGAI/SDR, MOEA/D, MOEA/D-DU, LMPFE, MOEA/D-UR, PREA, MCMOPSO-RL, DEAKP, and MOEA-2DE for 30 independent runs on the peak terrain set are summarized in Table 4. The Wilcoxon rank-sum test with a 95% confidence interval was used to test for significant differences. From the perspective of quantitative indicator results, MOEA-2DE has achieved considerable advantages in peak terrain sets and is superior to the comparison algorithms. The path curves of the compared algorithms on the peak terrain are shown in Figs. 3 and 4. Both figures show the frontal route and aerial views of each algorithm. The path curves can be preliminarily

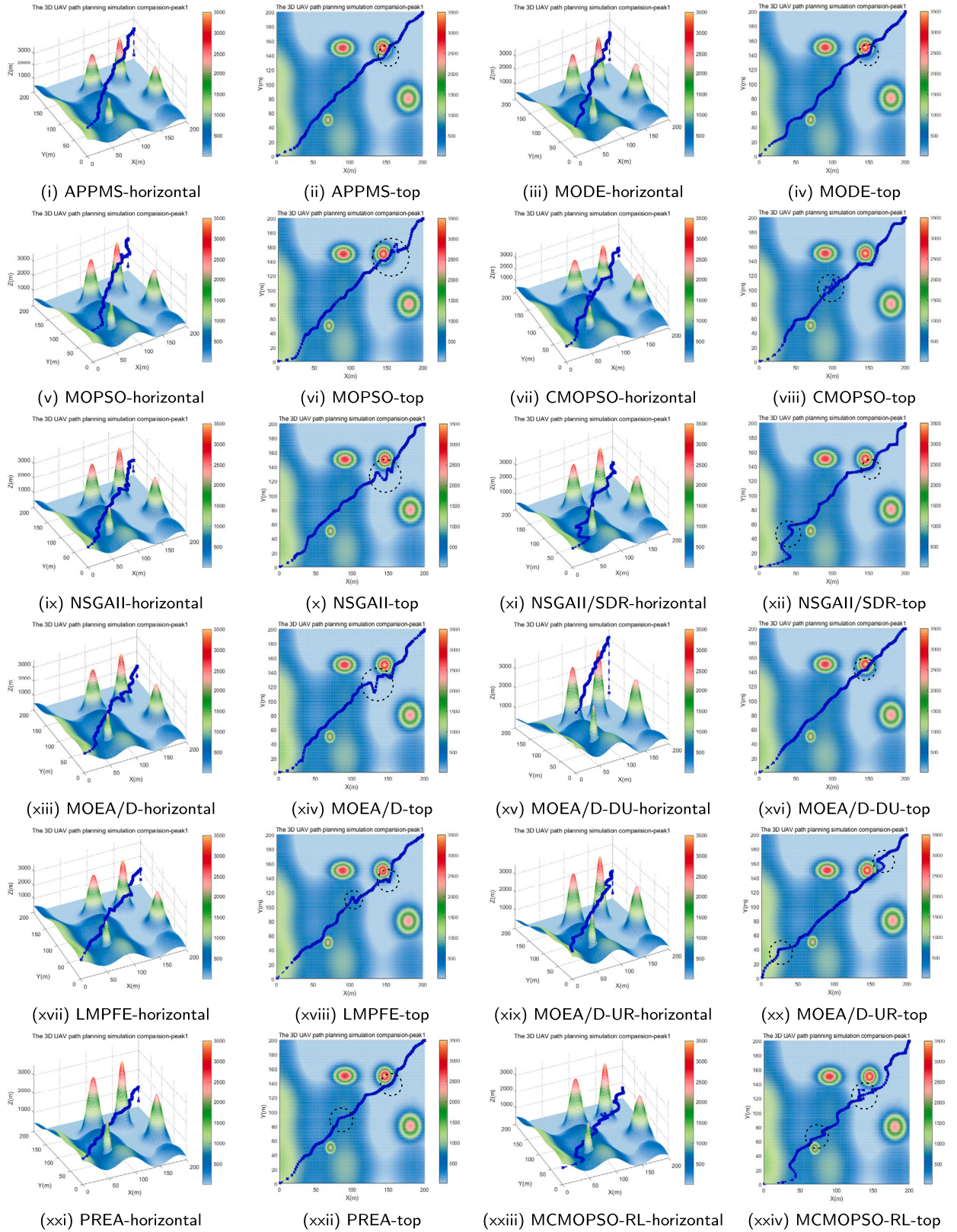


Fig. 3. The path curves of APPMS, MODE, MOPSO, CMOPSO, NSGAI, NSGAI/SDR, MOEA/D, MOEA/D-DU, LMPFE, MOEA/D-UR, PREA, MCMOPSO-RL, DEAKP and MOEA-2DE on the first peak terrain.

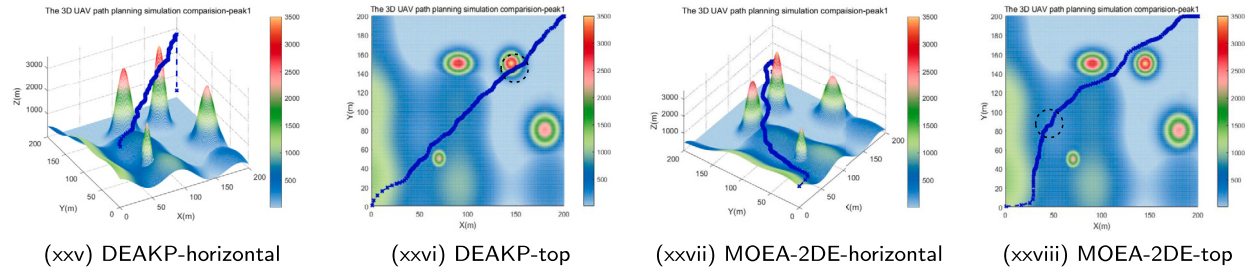


Fig. 3. (continued)

divided into two types of observation results: i) a straight path with a relatively high flight altitude, and ii) a relatively low flight altitude with a winding path.

On peak1 terrain, APPMS, MODE, MOPSO, MOEA/D-DU, and DEAKP were consistent with the situation described in (i). The results of the above algorithms yielded a shorter path through continuous climbing, thereby sacrificing the second goal. Meanwhile, the resulting path is relatively close to the peaks, as indicated by the black dotted circle in Fig. 3, which is not conducive to risk avoidance. Owing to their limited search capabilities, MODE and MOPSO cannot explore the desired area, resulting in poor performance. DEAKP is significantly affected by the knee point, and a knee point with poor performance has a sustained negative impact. The knee point in APPMS may have the same impact as that in DEAKP. MOEA/D-DU is less likely to obtain a more comprehensive PF in a low-dimensional objective space; thus, it is unable to leverage its advantages. However, MOEA/D-DU is capable of maintaining diversity (as suggested by the PD values in Table 4), which is consistent with the results of the study conducted by Yuan et al. [42] Examining the numerical results in Table 4, it is evident that the HV values of the above algorithms are low, which verifies that although the path length is short, an excessive flight altitude is not advisable. This also highlights the conflicting relationship between path length and flight altitude.

For condition (ii), the algorithms that match include CMOPSO, NSGAI, MOEA/D, NSGAI/SDR, LMPFE, MOEA/D-UR, PREA, MCMOPSO-RL, and MOEA-2DE. These algorithms can deviate from the shortest path and search for shorter paths at low flight altitudes, thereby demonstrating a compromised approach to the objective functions. As an advanced version of MOPSO, CMOPSO has achieved considerable results. However, it still faces challenges related to path oscillations caused by the aggregation of competitive particles. Although NSGAI and MOEA/D maintain a certain distance from the peaks and hold lower heights than previous algorithms, they exhibit a tortuous path and are still unable to explore a suitable route. As an improved algorithm for NSGAI, NSGAI/SDR achieves better results than its predecessor; however, it is still not ideal because the niching-technique-based method may provide solutions that are more unstable. PREA focuses on exploring selected promising regions that promote convergence to some extent; however, it may have the drawback of being trapped in local optima. The path curve obtained by LMPFE exhibited a few twists because of the difficulty in accurately capturing the estimated PF. MOEA/D-UR presented similar results, possibly because the distribution of the reference points may be uncertain, resulting in a biased weight setting. Intuitively, the path curve of the MCMOPSO-RL fluctuates significantly because of the diverse exploration modes that cause individual oscillations. However, this enhanced exploration ability allows for the exploration of paths at lower flight altitudes. Benefiting from the targeted search of the key dimension exploration strategy and the advantages of the adaptive evolution strategy, the proposed algorithm can rapidly search for promising solutions. Ignoring small-scale twists, MOEA-2DE exhibits excellent performance in terms of both the height and smoothness of the route.

In the more complicated peak problem, peak2, depicted in Fig. 4, all algorithms manifested varying degrees of twisting. According to the classification criteria mentioned in the previous experimental results, APPMS, MODE, MOPSO, DEAKP, and MOEA/D-DU maintained high flight heights corresponding to (i); therefore, promising solutions were not obtained. Specifically, APPMS, DEAKP, and MOEA/D-DU approach multiple peaks at high flight altitudes, resulting in a non-ideal route. MOPSO and MODE have weaknesses similar to those of the above algorithms; however, the path generated by MODE appears to be visually smoother. The numerical results of these algorithms were generally worse than those of other algorithms. In terms of criterion (ii), the remaining algorithms exhibited relatively low flight heights. However, exploration difficulties have also increased owing to the increasingly complex terrain. CMOPSO achieves a lower flight altitude, resulting in improved performance; however, it still shares the same flaws as peak1. Both NSGAI and MOEA/D showed more tortuous routes than the curves in peak1, whereas MOEA/D was smoother than NSGAI, as is evident from the numerical results. Similarly, PREA and LMPFE achieved lower flight altitudes, with LMPFE displaying better smoothness. The reason for the increased roughness in the NSGAI/SDR path is that the algorithm aims to explore lower-altitude positions but struggles to maintain stable solutions. However, the path curve of MCMOPSO-RL is comparatively smooth owing to its updated cumulative performance mode. However, defects still exist near the mountain peak. In terms of the comparison of the PD values and the significance of the rank sum test, the proposed algorithm performed similarly to MCMOPSO-RL in terms of the diversity indicator; however, it remained competitive when combined with the HV indicator. Overall, when faced with more complex peak problems, the proposed algorithm provides a smoother and safer path than the other algorithms.

The average convergence traces of all algorithms over 30 independent runs on the peak terrains are shown in Fig. 5. The upper part of Fig. 5 shows the convergence curve for the HV values and the lower part displays the convergence curve for the PD values. Notably, MOEA-2DE achieved the best final convergence results in most cases, except for the PD value in the peak2 terrain. With

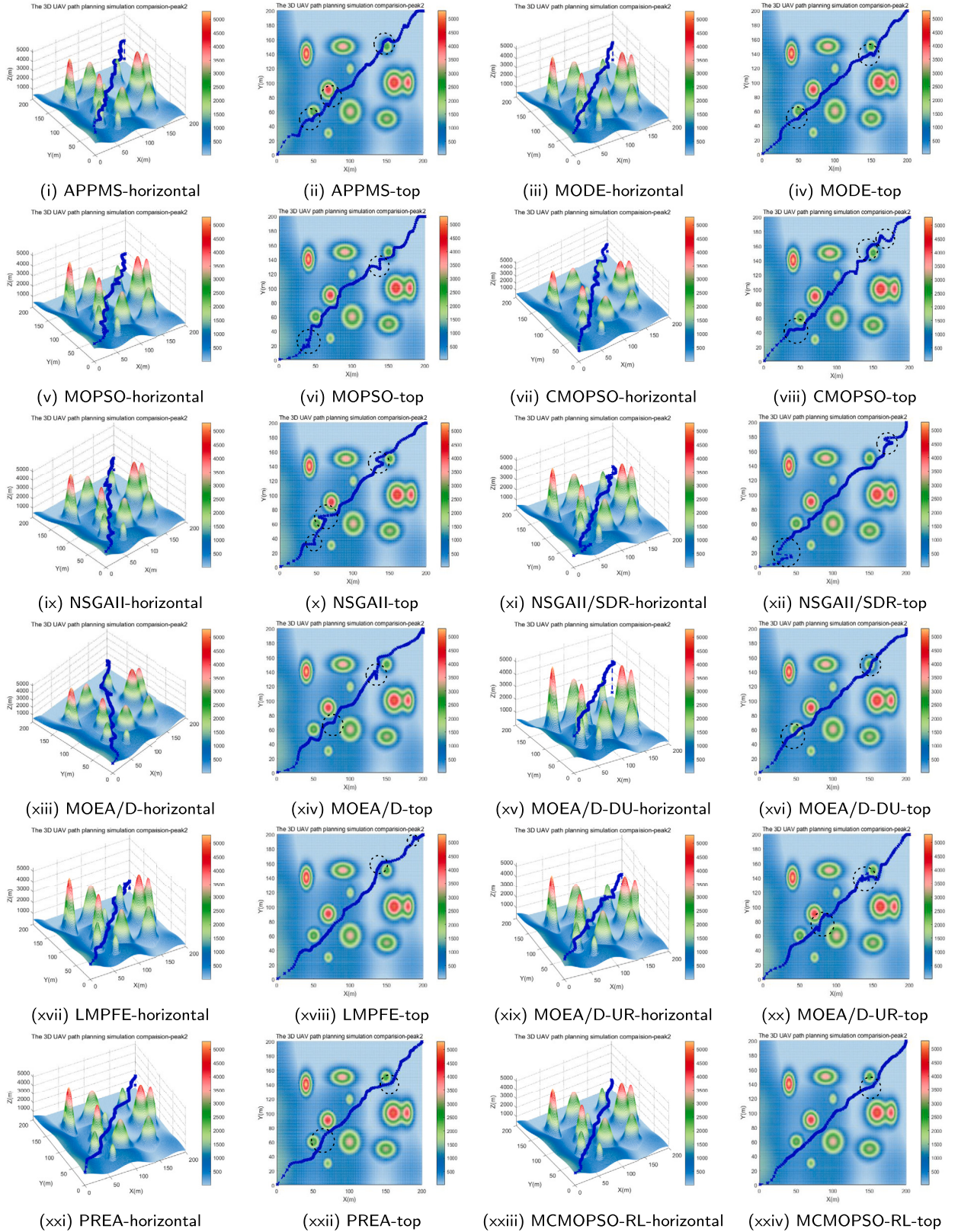


Fig. 4. The path curve of APPMS, MODE, MOPSO, CMOPSO, NSGAI, NSGAI/SDR, MOEA/D, MOEA/D-DU, LMPFE, MOEA/D-UR, PREA, MCMOPSO-RL, DEAKP and MOEA-2DE on the second peak terrain.

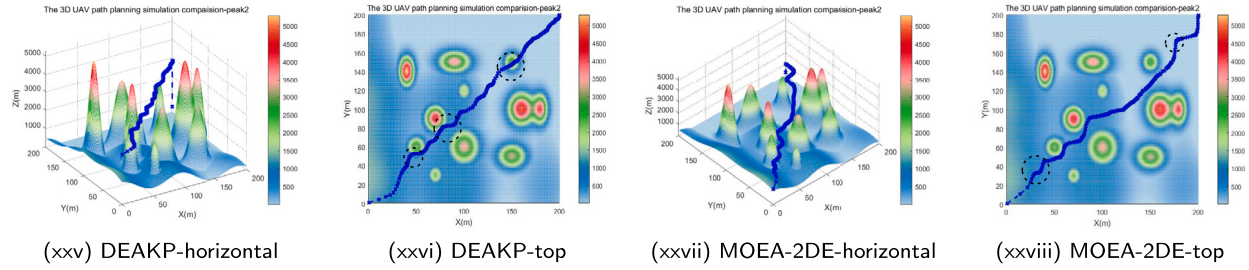


Fig. 4. (continued)

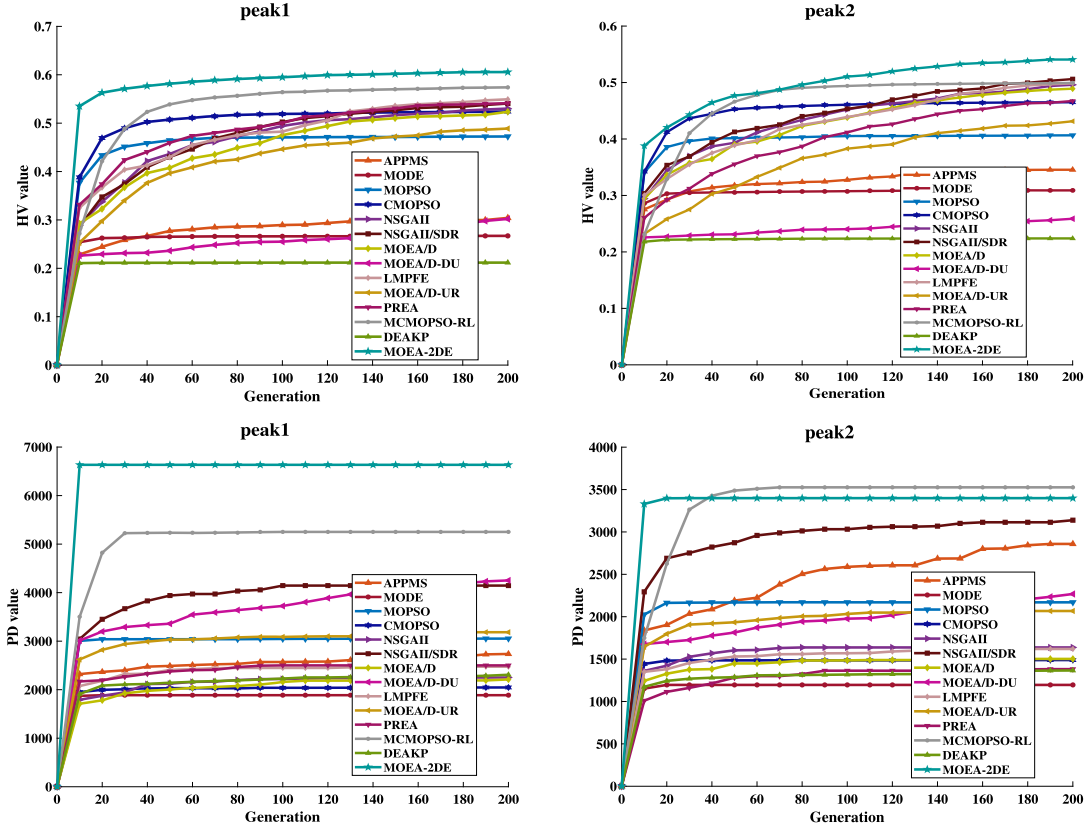


Fig. 5. The average convergence traces of APPMS, MODE, MOPSO, CMOPSO, NSGAI, NSGAI/SDR, MOEA/D, MOEA/D-DU, LMPFE, MOEA/D-UR, PREA, MCMOPSO-RL, DEAKP and MOEA-2DE on the peak terrains.

regard to the convergence speed, the proposed algorithm also outperformed other comparative algorithms because of the adaptive adjustment of strategies to the evolutionary process. In conclusion, MOEA-2DE has achieved preferable performance on peak terrain.

4.4. Results on rugged terrain scene

Table 5 tabulates the average HV value, PD value, and corresponding standard deviation obtained by all the compared algorithms over 30 independent runs on rugged terrain. A Wilcoxon rank-sum test with a 95% confidence level was also conducted, and the best results are highlighted in boldface. As shown in Table 5, MOEA-2DE demonstrates superiority over all comparison algorithms for rugged terrain test problems. The path curves of all the algorithms on rugged terrain are presented in Figs. 6 and 7. The height distribution of rugged terrain problems varies significantly, requiring an algorithm to achieve a sustainable balance between shorter paths and lower altitudes. In this part of the experiment, the performances of the algorithms were classified and analyzed based on their tendency to span high altitudes. Similarly, it is divided into two categories: (i) crossing high altitudes multiple times to obtain short paths but sacrificing height and causing performance degradation, and (ii) attempting to find low-altitude zones to achieve a compromise between low altitude and short paths, examining the search ability of the algorithm.

From the perspective of rugged1, the algorithms consistent with situation (i) included APPMS, MOPSO, MOEA/D, MOEA/D-DU, MOEA/D-UR, and DEAKP. Most algorithms find it difficult to avoid using high-altitude paths, partly because of their limited search

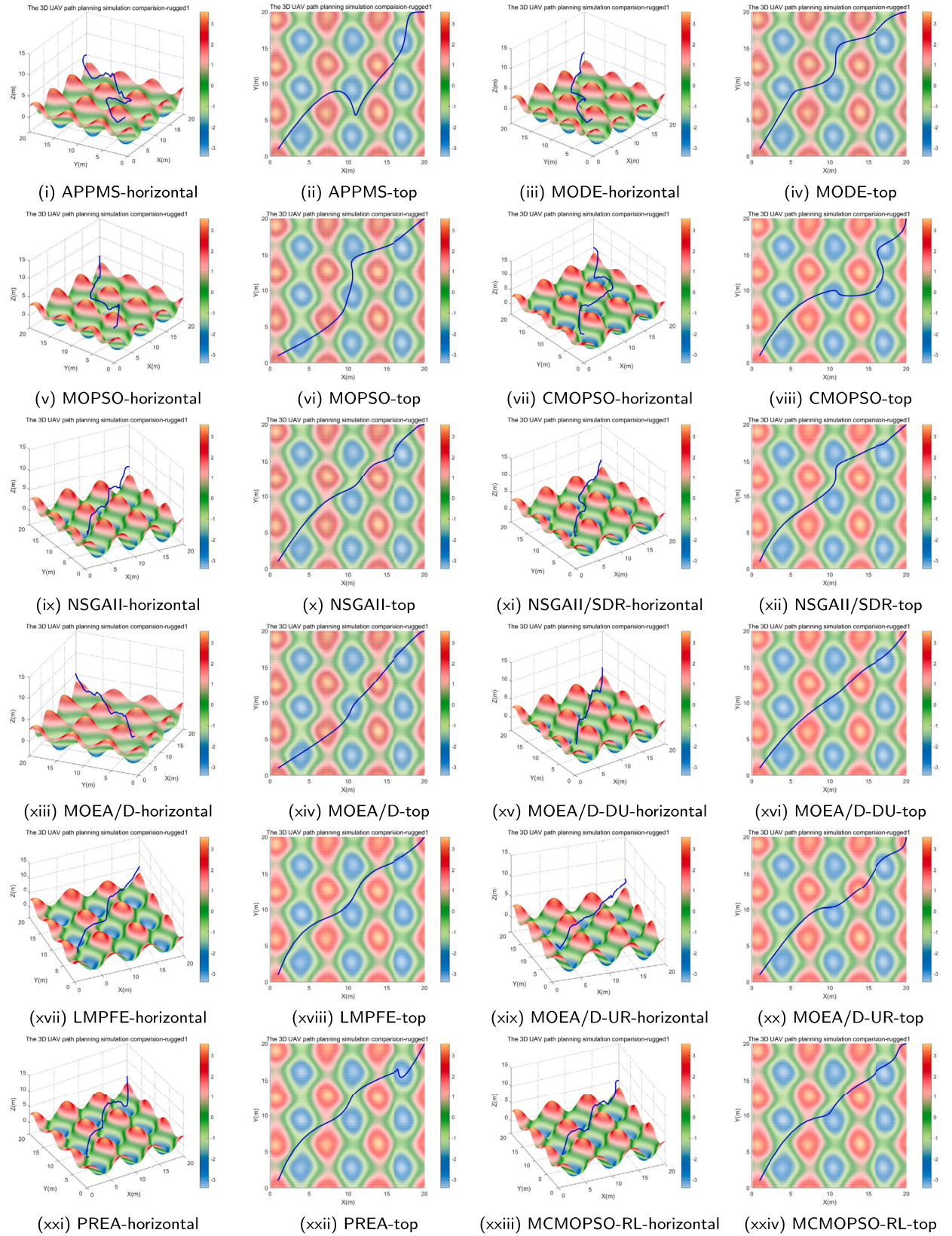


Fig. 6. The path curve of APPMS, MODE, MOPSO, CMOPSO, NSGAI, NSGAI/SDR, MOEA/D, MOEA/D-DU, LMPFE, MOEA/D-UR, PREA, MCMOPSO-RL, DEAKP and MOEA-2DE on the first rugged terrain.

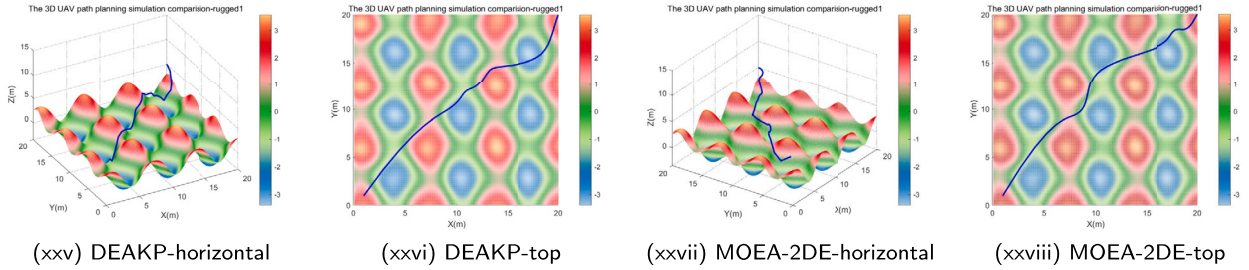


Fig. 6. (continued)

Table 5

Averaged HV value, PD value and corresponding standard deviations obtained by APPMS, MODE, MOPSO, CMOPSO, NSGAI, NSGAI/SDR, MOEA/D, MOEA/D-DU, LMPFE, MOEA/D-UR, PREA, MCMOPSO-RL, DEAKP and MOEA-2DE on rugged terrains.

Terrain	Metrics	rugged1		rugged2	
		HV	PD	HV	PD
APPMS		2.6787E-01(+) $\pm 8.39E-03$	5.0445E+01(+) $\pm 4.66E+01$	1.1649E-01(+) $\pm 6.84E-03$	4.7224E+01(+) $\pm 4.67E+01$
MOPSO		3.1447E-01(+) $\pm 1.01E-02$	2.6493E+01(+) $\pm 2.47E+01$	1.6319E-01(+) $\pm 1.81E-02$	5.2850E+01(+) $\pm 5.40E+01$
CMOPSO		3.3333E-01($\approx$) $\pm 9.80E-03$	3.1569E+01(+) $\pm 2.65E+01$	1.9503E-01(+) $\pm 2.69E-02$	6.1053E+01(+) $\pm 4.44E+01$
NSGAI		3.3495E-01($\approx$) $\pm 8.70E-03$	2.9744E+01(+) $\pm 2.09E+01$	2.0941E-01(+) $\pm 1.34E-02$	5.8214E+01(+) $\pm 3.86E+01$
NSGAI/SDR		3.3082E-01(+) $\pm 8.63E-03$	2.9368E+01(+) $\pm 1.94E+01$	2.0854E-01(+) $\pm 1.51E-02$	3.8065E+01(+) $\pm 2.41E+01$
MOEA/D		3.2300E-01(+) $\pm 1.06E-02$	3.0077E+01(+) $\pm 2.18E+01$	1.8250E-01(+) $\pm 1.57E-02$	3.6181E+01(+) $\pm 2.79E+01$
MOEA/D-DU		2.8207E-01(+) $\pm 1.03E-02$	1.6057E+01(+) $\pm 9.64E+00$	1.5430E-01(+) $\pm 1.23E-02$	2.7030E+01(+) $\pm 1.98E+01$
MODE		3.2482E-01(+) $\pm 1.28E-02$	2.1778E+01(+) $\pm 2.00E+01$	1.7882E-01(+) $\pm 3.27E-02$	3.2869E+01(+) $\pm 2.79E+01$
LMPFE		3.3365E-01($\approx$) $\pm 8.04E-03$	2.6725E+01(+) $\pm 1.74E+01$	2.1399E-01(+) $\pm 1.55E-02$	5.0484E+01(+) $\pm 3.92E+01$
MOEA/D-UR		3.0674E-01(+) $\pm 8.98E-03$	1.6995E+01(+) $\pm 1.75E+01$	1.5835E-01(+) $\pm 2.23E-02$	2.2160E+01(+) $\pm 1.89E+01$
PREA		3.3307E-01($\approx$) $\pm 1.01E-02$	1.9903E+01(+) $\pm 1.41E+01$	2.0790E-01(+) $\pm 1.61E-02$	3.9198E+01(+) $\pm 3.11E+01$
MCMOPSO-RL		3.1375E-01(+) $\pm 7.88E-03$	4.9423E+01(+) $\pm 2.91E+01$	1.6305E-01(+) $\pm 1.73E-02$	7.5245E+01(+) $\pm 3.21E+01$
DEAKP		2.9772E-01(+) $\pm 9.65E-03$	3.9826E+01(+) $\pm 2.51E+01$	1.3378E-01(+) $\pm 1.02E-02$	6.4756E+01(+) $\pm 2.49E+01$
MOEA-2DE		3.3525E-01 $\pm 8.13E-03$	8.3840E+01 $\pm 3.51E+01$	2.2982E-01 $\pm 1.70E-02$	1.2336E+02 $\pm 3.84E+01$

ability and the attraction of the shortest path, which leads to local optima. APPMS tends to find low terrain; however, the path is relatively deviated and cannot achieve the desired effect. MOPSO, MOEA/D, and MOEA/D-DU seek shorter paths by climbing over partially high points, leading to decreased performance. Although the paths of MOEA/D and MOEA/D-DU appeared to be the shortest from an aerial perspective, increasing the height increased the cost, causing an unexpected effect. MOEA/D-UR and DEAKP showed similar trends; however, DEAKP provided better diversity based on PD values (Table 5).

Other algorithms can be classified under category (ii), which explore additional possibilities at low flight altitudes. NSGAI, MODE, and NSGAI/SDR can avoid directly crossing high points and seeking lower terrain but still exhibit significant curve changes. Although the path of CMOPSO appears longer, most of the path points pass through the lower parts, resulting in improved performance. LMPFE and PREA also have a positive effect owing to the effective estimation of promising regions. Owing to the search ability of the key dimension exploration strategy for special regions, the proposed algorithm achieved the best results. MOEA-2DE not only provides a path with lower terrain but also provides a path with higher smoothness, demonstrating a desirable result.

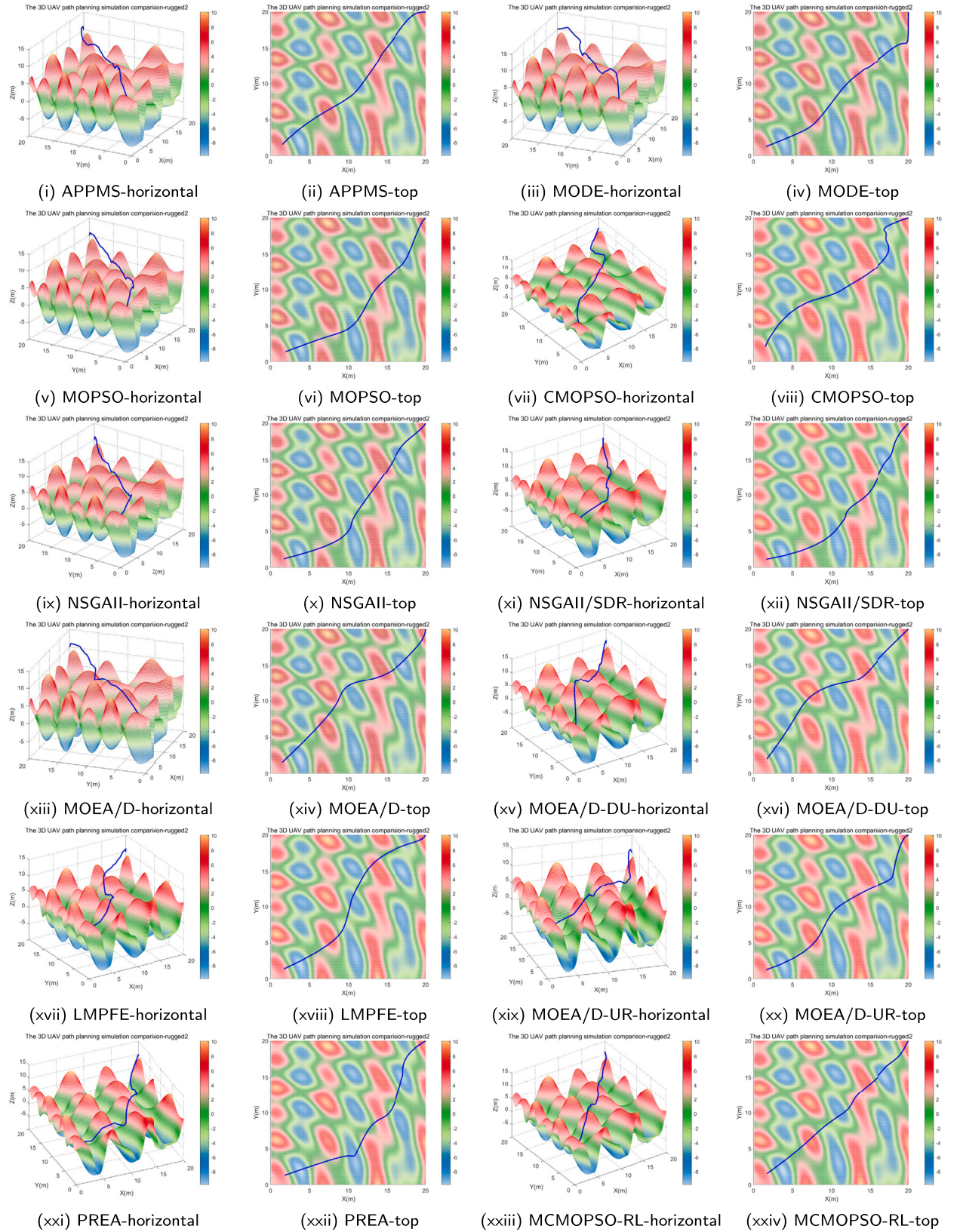


Fig. 7. The path curve of APPMS, MODE, MOPSO, CMOPSO, NSGAI, NSGAI/SDR, MOEA/D, MOEA/D-DU, LMPFE, MOEA/D-UR, PREA, MCMOPSO-RL, DEAKP and MOEA-2DE on the second rugged terrain.

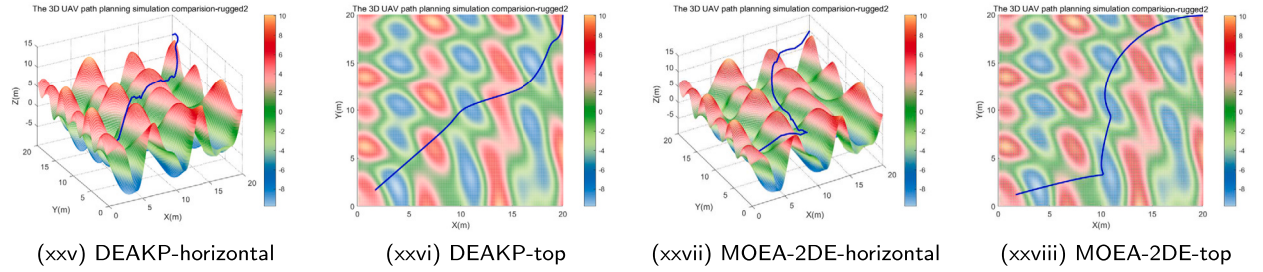


Fig. 7. (continued)

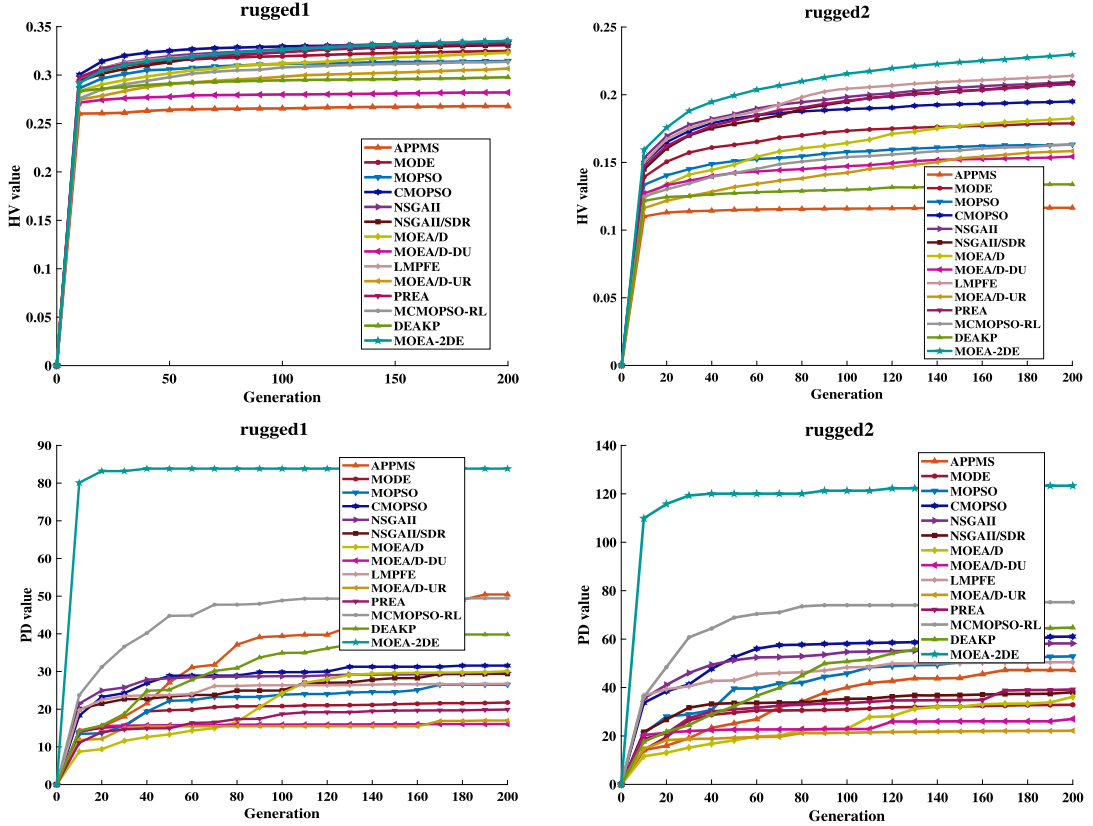


Fig. 8. The average convergence traces of APPMS, MODE, MOPSO, CMOPSO, NSGAI, NSGAI/SDR, MOEA/D, MOEA/D-DU, LMPFE, MOEA/D-UR, PREA, MCMOPSO-RL, DEAKP and MOEA-2DE on the rugged terrains.

In comparison, rugged2 presents a more challenging terrain with an uneven distribution, which explains the higher requirements for the algorithm's exploration ability and the greater difficulty in obtaining short paths at low flight altitudes. As depicted in Fig. 7, almost all algorithms inevitably increase the flight altitude because the difficulty of searching increases significantly. APPMS and MODE directly span multiple high points, and the paths obtained are rarely smooth. MOPSO, NSGAI, and MOEA/D adjust their directions to achieve lower flight altitudes; however, they have not yet achieved a good compromise solution. Their subsequent algorithms, the CMOPSO, NSGAI/SDR, MOEA/D-UR, and MOEA/D-DU, experienced the same issues owing to their insufficient search capabilities. Despite exploring a certain number of low-altitude routes, LMPFE and PREA remained trapped at a few peaks. MCMOPSO-RL appears to have a relatively straight path; however, its performance declines because of the cost across multiple peaks. Compared with other algorithms, the proposed algorithm generates a route that does not require crossing peaks, demonstrating strong search performance and verifying the effectiveness of the targeted search. Although MOEA-2DE may not provide the shortest path, it can capture a compromised path at a lower height, thereby satisfying the expected path requirements.

The average convergence traces of all algorithms over 30 independent runs on rugged terrain are shown in Fig. 8. The differences between the HV values in peak1 were not significant; however, the best result was obtained using MOEA-2DE. Additionally, the PD values of the proposed algorithm for rugged problems were significantly better than those of the other algorithms, indicating its

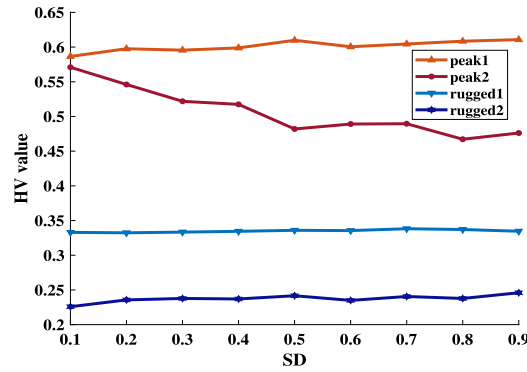


Fig. 9. The result of MOEA-2DE with different SD values after running 30 times independently on all problems.

Table 6

The running time of all comparison algorithms after running 30 times independently on selected problems, in seconds, accurate to two decimal places.

APPMS	NSGAI	MOEA/D	MOPSO	MODE	CMOPSO	NSGAI/SDR
26.61/5.81	20.89/3.32	22.61/3.64	19.16/3.21	22.98/3.49	19.95/3.61	24.28/3.73
MOEA/D-DU	LMPFE	MOEA/D-UR	PREA	MCMOPSO-RL	DEAKP	MOEA-2DE
33.11/4.51	32.72/12.98	24.55/4.78	20.80/3.17	18.98/3.42	34.35/4.91	19.65/3.21

ability to maintain diversity. MOEA-2DE achieved superior performance on other problems and metrics, both in terms of convergence speed and final convergence results, further confirming the efficacy of the strategies on both simple and complex problems.

4.5. Sensitivity analysis of the parameter

This section discusses the impact of a parameter in the proposed algorithm, specifically the mutation probability SD of dimensions other than the key dimensions. The numerical value of SD was set to $[0.1, 0.9]$, incremented by 0.1. A testing experiment was conducted on all four problems using different SD values, and the results are shown in Fig. 9. The figure shows that varying SD values exhibited relatively stable performance on issues peak1, rugged1, and rugged2. A larger SD value can cause significant changes in the decision space. However, the adaptive evolution strategy based on population discrepancy uses KLD to detect differences, thereby ensuring diversity in the offspring generation strategies. This effect was maintained when it was combined with environmental selection. However, when it comes to issue peak2, increasing the SD value results in a decrease in performance owing to the complex terrain and substantial changes in the decision space. An increase in the SD value disables the targeted search to a certain extent, thereby leading to performance degradation. In conclusion, $SD = 0.3$ is appropriate as an unbiased choice in this study.

4.6. The comparison of running time

A time-complexity analysis of the proposed algorithm is presented in Section 3.2.4. To further analyze the time performance of the algorithms, the running times of all comparison algorithms were investigated. Two relatively sophisticated problems, namely peak2 and rugged2, were chosen for testing. Table 6 lists the running time results of all algorithms on the selected problems. Each value represents the average of 30 independent runs, and the best results are shown in bold. Because the search space of the peak problem is larger than that of the rugged problem, the overall time required is longer. With respect to the results on the peak problem, the running time ranges from 18.98 s to 34.35 s. The running time of MOEA-2DE is only 0.67 s away from the minimum value, and the superiority of the proposed algorithm makes such a time difference worth it. From the results of the rugged problem, the running time varies between 3.17 s and 12.98 s. With a smaller search space, the running time differences between most algorithms are relatively small (the longer time of LMPFE can be attributed to the complexity of its environment selection and fitness calculation). The running time of the proposed algorithm approached the minimum time required. Overall, MOEA-2DE can achieve a good balance between performance and complexity in terms of overall effectiveness and obtain a better solution within an acceptable time cost.

5. Conclusion and future work

To solve the UAV path-planning problem in complex environments more effectively, this study proposes a multi-objective optimization-based UAV path-planning model and designs a MOEA based on dimension exploration and discrepancy evolution, namely, MOEA-2DE. Specifically, the MOEA-2DE is characterized by two strategies. The key dimension exploration strategy employs

dimension perturbations to identify key dimensions, achieve a prior exploration of issues, and lay the foundation for subsequent targeted exploration. The adaptive evolution strategy based on population discrepancy applies KLD to assess differences in population distribution throughout the evolution process and adaptively selects different offspring generation methods to promote convergence or balance diversity in response to changes during evolution. As expected, MOEA-2DE outperformed the various comparison algorithms on the designed complex terrain sets.

Although the proposed algorithm achieved promising results, there were still a few limitations that deserve further investigation. For example, *SD* requires manual adjustment for different issues, and adopting an adaptive method to determine *SD* is more desirable. Additionally, when using KLD to measure population discrepancies, recent evolutionary trends may be biased, affecting the accuracy of the population distribution. More historical population experience can reflect the tendency of the population distribution more accurately. From a practical perspective, the model should implement more UAV parameters such as speed, acceleration, and attitude. In future work, real-terrain data will be used to test the effectiveness of the proposed algorithm.

Despite these limitations, the findings of this study are significant, as they could aid unmanned aerial vehicles in successfully completing complex missions while conserving time and resources, which could considerably advance military and civil aerial exploration.

CRedit authorship contribution statement

Xiuju Xu: Methodology, Formal analysis, Software, Writing – original draft. **Chengyu Xie:** Software, Validation. **Zongfu Luo:** Review. **Chuanfu Zhang:** Review. **Tao Zhang:** Review, Editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code is available at <https://github.com/Xuxiuju-sysu/MOEA-2DE>.

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